INCH POUND
MIL-PRF-1/385D
1 October 2009
SUPERSEDING
MIL-PRF-1/385B
7 April 1999

PERFORMANCE SPECIFICATION SHEET

# ELECTRON TUBE, CATHODE RAY
# TYPES 3DP1A AND 3DP1A+S2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Electrostatic deflection and focus with decalcomania inscription on tube type 3DP1A+S2

DIMENSIONS AND PIN CONNECTIONS: See figures 1 and 2

ABSOLUTE RATINGS:

| Parameter: | Ef | Ec1 | ed | Eb1 | Eb2 | Light output | Rg | Zd | Ehk | Alt |
|---|---|---|---|---|---|---|---|---|---|---|
| Unit: | V | V dc | v | V dc | V dc | fL | MegΩ | MegΩ | V dc | ft |
| Maximum: | 6.9 | 0 | 550 | 1,100 | 2,200 | - - - | 1.5 | 1.0 | -125 | 10,000 |
| Minimum: | 5.7 | -200 | - - - | - - - | 1,500 | 7.5 | - - - | - - - | - - - | - - - |
| Test Conditions: | 6.3 | Adj | - - - | Focus | 1,500 | - - - | - - - | - - - | - - - | - - - |

GENERAL:

Qualification - Required



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:18Z
Check the source to verify that this is the current version before use.

TABLE 1. Testing and inspection.

| Inspection | Method MIL-STD-1311 | Notes | Conditions | Symbol | Limits Min | Limits Max | Units |
|---|---|---|---|---|---|---|---|
| Qualification inspection | | | | | | | |
| Neck and bulb alignment (electrostatic types) | 5101 | - | | Diam | - - - | 2.25 | Inches |
| Cathode illumination | 5216 | - | | - - - | - - - | - - - | - - - |
| Deflection-factor uniformity | 5248 | - | | - - - | - - - | - - - | - - - |
| Direct-interelectrode capacitance | 1331 | - | Control grid to all | Cg1 | - - - | 10.5 | pF |
| | | | Cathode to all | Ck | - - - | 10.5 | pF |
| | | | D1 to D2 | C1D2 | - - - | 4.0 | pF |
| | | | D3 to D4 | C3D4 | - - - | 4.0 | pF |
| | | | D1 to all | CD1 | - - - | 15.0 | pF |
| | | | D3 to all | CD3 | - - - | 12.0 | pF |
| | | | D1 to all, except D2 | CD1 | - - - | 13.0 | pF |
| | | | D2 to all, except D1 | CD2 | - - - | 13.0 | pF |
| | | | D3 to all, except D4 | CD3 | - - - | 12.0 | pF |
| | | | D4 to all, except D3 | CD4 | - - - | 12.0 | pF |
| | | | D5 to anode 2 | CD5 | - - - | 3.5 | pF |
| Base material insulating quality | 1216 | - | Zone 5 (min) | - - - | - - - | - - - | - - - |
| Pressure (implosion) | 1141 | - | | - - - | - - - | - - - | - - - |
| Conformance inspection, part 1 | | | | | | | |
| Voltage breakdown | 5201 | - | | - - - | - - - | - - - | - - - |
| Voltage breakdown (electrostatic types) | 5201 | - | | - - - | - - - | - - - | - - - |
| Gas “cross” | 5206 | 1, 2 | Light = 7.5 fL | - - - | - - - | - - - | - - - |
| Bulb, screen and faceplate quality | 5106 | - | | - - - | - - - | - - - | - - - |
| Light output | 5221 | - | | Light | 7.5 | - - - | fL |
| Modulation | 5223 | - | Light = 7.5 fL | ΔEc | - - - | 40 | V dc |
| Spot position (electrostatic deflection) | 5231 | - | | - - - | - - - | 15 | mm |
| Spot displacement (leakage) | 5231 | - | | Displ | - - - | 5 | mm |
| Grid cutoff voltage | 5241 | - | | Ecl | -22.5 | -67.5 | V dc |
| Grid No. 1 leakage current | 5251 | - | | - - - | - - - | - - - | - - - |
| Anode No. 2 leakage current | 5251 | - | | - - - | - - - | - - - | - - - |



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:18Z
Check the source to verify that this is the current version before use.

TABLE 1. Group A inspection -Continued.

| Inspection | Method MIL-STD-1311 | Notes | Conditions | Symbol | Limits Min | Limits Max | Units |
|---|---|---|---|---|---|---|---|
| Conformance inspection, part 2 | | | | | | | |
| Heater current | 1301 | - | | If | 540 | 660 | mA |
| Electrode current (anode No. 1) | 5201 | - | Ec1 = 0 | Ib1 | -50 | 10 | µA dc |
| Electrode current (cathode) | 5201 | - | Light = 7.5 fL | Ik | - - - | 1,000 | µA dc |
| Base alignment (electrostatic types) | 5101 | - | +1D2, pin No. 5 | - - - | - - - | - - - | - - - |
| Angle between traces | 5101 | - | | - - - | - - - | - - - | - - - |
| Neck and base alignment (electrostatic types) | 5101 | - | | - - - | - - - | - - - | - - - |
| Stray light emission (conventional types) | 5216 | - | Eb2 = 2,200 V dc | - - - | - - - | - - - | - - - |
| Line width “A” (electrostatic deflection) | 5226 | 3, 5 | Light = 7.5 fL | Width | - - - | 0.60 | mm |
| Line width “B” (electrostatic deflection) | 5226 | 3, 6 | Light = 7.5 fL | Width | - - - | 0.80 | mm |
| Focusing voltage at cutoff | 5246 | - | | Eb1 | 345 | 515 | V dc |
| Focusing voltage (zero bias) | 5246 | - | | Eb1 | 300 | 515 | V dc |
| Deflection factor | 5248 | - | 1D2 | DF | 127 | 173 | V dc/in. |
| Deflection factor | 5248 | - | 3D4 | DF | As recorded | As recorded | V dc/in. |
| Deflection factor | - - - | 4 | D5 | Pk to Pk | 50 | 84 | V dc |
| Radial pattern width | - - - | 4 | | Width | - - - | 10 | mm |
| Deflection ratio | - - - | - | 1D2/3D4 | Ratio | 1.14 | 1.56 | - - - |
| Heater-cathode leakage current | 5251 | - | | - - - | - - - | - - - | - - - |
| Secureness of base, cap or insert | 1101 | - | | - - - | - - - | - - - | - - - |
| Base pin solder depth | 1111 | - | | - - - | - - - | - - - | - - - |
| Permanence of marking | 1105 | - | | - - - | - - - | - - - | - - - |



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:18Z
Check the source to verify that this is the current version before use.

TABLE 1. Testing and inspection -Continued.

| Inspection | Method MIL-STD-1311 | Notes | Conditions | Symbol | Limits Min | Limits Max | Units |
|---|---|---|---|---|---|---|---|
| Conformance inspection, part 3 | | | | | | | |
| Life test | - - - | - | Group C; Light = 7.5 fL; Eb2 = 2,200 V dc; t = 500 hours | - - - | - - - | - - - | - - - |
| Life-test end points: | - - - | | | | | | |
| Modulation | 5223 | - | Light = 7.5 fL | ΔEc | - - - | 50 | V dc |
| Line width “A” | 5226 | - | Light = 7.5 fL | Width | - - - | 0.70 | mm |
| Line width “B” | 5226 | - | Light = 7.5 fL | Width | - - - | 1.05 | mm |

NOTES:

1. The center of the “gas cross” appearing inside pattern will not appear due to presence of rod in tube face.
2. This test to be performed at the conclusion of the holding period.
3. Modify the regular procedure of line width measurements in the following manner: Adjustment shall be made for best focus at a point ½ inch (12.70 mm) from the center of the radial deflecting electrode terminal along the line of the high-frequency scan.
4. Using mask centered with respect to tube face, set diameter of scanned circle to 1.375 inch (34.93 mm) and remove mask. Apply and adjust 35 KHz signal on D5 (rod) so that minimum width of annular pattern is 5 mm. The maximum radial width of same pattern shall be within the limit specified. The peak voltage shall be within the limits specified.
5. The tubes shall be observed for deflection defocusing, astigmatism and spot ellipticity observable to the eye, as evidenced by fuzziness due to lack of sharpness of trace (usually around the edges), bow-tying (irregular widths of any line when observed at different points) and bowing of trace other than that normally caused by curvatures of the bulb.
6. An examination for defocusing, astigmatism, or spot ellipticity shall be made as in note 5.
7. The following applies to tube type 3DP1A+S2.

   (a) Before application of the inscription, the tube shall have passed all requirements of this specification sheet.

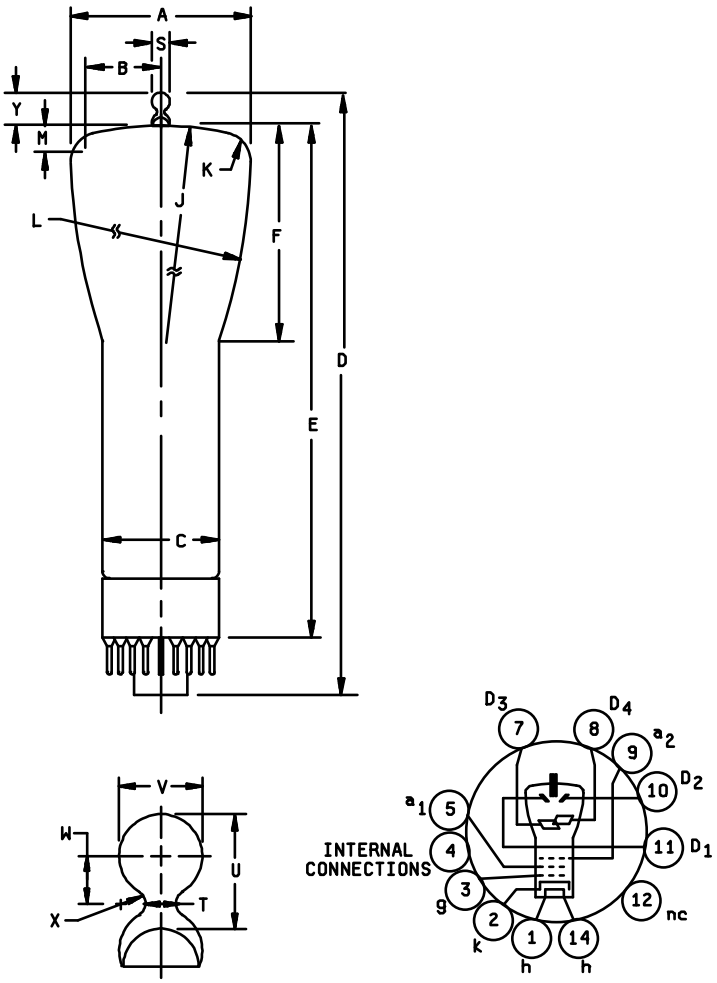
   (b) The inscription outlined on figure 2 shall be secured to the tube and be positioned in the following manner:

   (1) The trace produced by deflecting electrodes 1D2 shall not deviate by more than 2.5 degrees from a line passing through scale markings 0 and 2.5 on the inscription.
   (2) The position of zero on the inscription shall be 64.28 degrees ±12.5 degrees clockwise from the centerline of the base lug key, looking at the tube face.
   (3) The inscription shall be centered with respect to the tube face within .047 (1.19 mm) of an inch.
   (4) Mask center electrode.
   (5) After masking, the tube face shall be sprayed with a light coat of clear lacquer.

8. This specification sheet uses accept on zero defect sampling in accordance with MIL-PRF-1, table III.



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:18Z
Check the source to verify that this is the current version before use.




| Dimensions | | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| Ltr | Min | Max | Min | Max |
| Conformance inspection, part 2 | | | | |
| A | 2.937 | 3.063 | 74.60 | 77.80 |
| B | See note 1 | | | |
| C | 1.937 | 2.063 | 49.20 | 52.40 |
| D | 10.063 | 10.750 | 255.60 | 273.08 |
| S | See note 2 | | | |
| U | .141 | | 3.58 | |
| V | .110 | .116 | 2.79 | 2.95 |
| Y | .313 | .501 | 7.95 | 12.73 |
| Reference dimensions (see note 6) | | | | |
| E | 9.250 | | 234.95 | |
| F | 3.875 | | 98.43 | |
| J | 8.000 R | | 203.20 R | |
| K | .250 R | | 6.35 R | |
| L | 12.438 R | | 315.93 | |
| M | .350 | | 8.89 | |
| T | .068 Dia | | 1.73 Dia | |
| W | .058 | | 1.47 | |
| X | .030 R | | 0.76 R | |

NOTES:

1. The minimum useful screen radius shall not be less than 1.375 (34.93 mm)
2. The maximum diameter uncoated area around center of the radial deflecting electrode terminal shall not be more than .625 (15.88 mm).
3. The radial deflecting electrode terminal shall be eccentric with respect to the tube axis by not more than .063 (1.60 mm).
4. The base shall be a medium shell DIHEPTAL 12-pin (B12-37) type.
5. The bulb shall be a J24G type.
6. These dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube 3DP1A and 3DP1A+S2.



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:18Z
Check the source to verify that this is the current version before use.

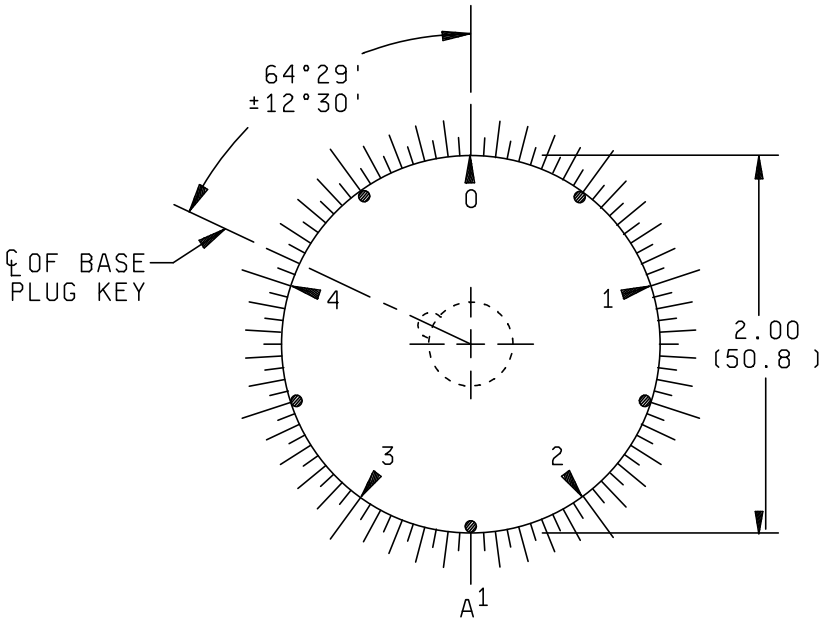



FIGURE 2. S2 scale.

Referenced documents. In addition to MIL-PRF-1, this document references the following:
MIL-STD-1311

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Preparing activity:
DLA - CC

(Project 5960-2009-009)

Review activities:
Navy - AS, CG, MC, OS
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:18Z
Check the source to verify that this is the current version before use.